

BAYESIAN DECISION AND UNCERTAINTY: SKIN LESION TRIAGE

Classical Pipeline | TU Berlin Summer Semester 2026

Group: 01

Zohreh Asadi

Jannis Ole Röhl

Fenya Waßerroth

David Celik

PROJECT OVERVIEW AND DATASET



- Binary classification task: melanoma vs. benign lesions
- Includes calibrated uncertainty estimation
- Dataset: HAM10000 dermoscopy images
- Class imbalance:
 - 6,231 benign samples
 - 779 melanoma cases in training set



- Medical triage prioritizes minimizing false negatives due to higher risks.
- Feature extraction uses HSV color space histograms.
- Captures key visual patterns from dermoscopy images.

ASYMMETRIC THRESHOLD & MEDICAL TRIAGE IMPLICATIONS

- Lower threshold (0.091) increases melanoma detection sensitivity
 - Melanoma predictions increase from 56 to 60 cases
- Asymmetric triage metrics:
 - Melanoma recall improves: 0.13 vs 0.12
 - Precision remains stable at 0.23
 - Benign metrics unchanged
- Key trade-off: Enhanced safety for melanoma detection at cost of more false positives
- Bayesian decision boundaries set thresholds based on asymmetric cost matrix

CLASSICAL PIPELINE & MODEL SELECTION



Convert dermoscopy images to HSV color space and compute concatenated per-channel histograms as feature vectors for



Train Gaussian Mixture Models (GMMs) per class using Expectation-Maximization algorithm, testing $K=1$ to 10 components.



Select optimal K by minimizing Bayesian Information Criterion (BIC), then apply Bayesian decision theory with asymmetric cost matrix.

```
Number of benign samples for training: 6231
Number of melanoma samples for training: 779
Training arc

Training GMM for benign samples with K=1 components...
Training GMM for melanoma samples with K=1 components...

Training GMM for benign samples with K=2 components...
Training GMM for melanoma samples with K=2 components...

Training GMM for benign samples with K=3 components...
Training GMM for melanoma samples with K=3 components...

Training GMM for benign samples with K=4 components...
Training GMM for melanoma samples with K=4 components...

Training GMM for benign samples with K=5 components...
Training GMM for melanoma samples with K=5 components...

Training GMM for benign samples with K=6 components...
Training GMM for melanoma samples with K=6 components...

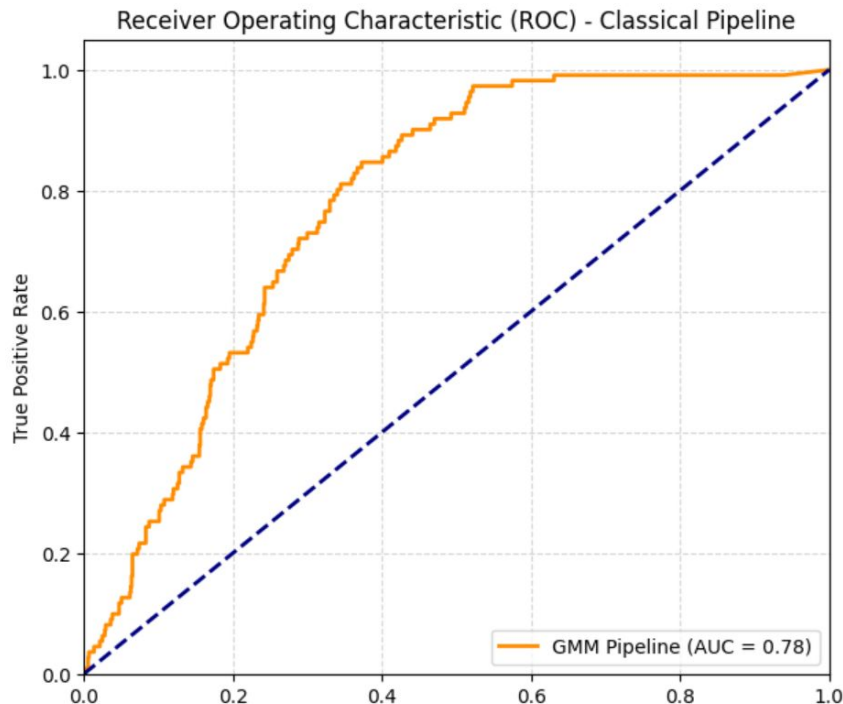
Training GMM for benign samples with K=7 components...
Training GMM for melanoma samples with K=7 components...

Training GMM for benign samples with K=8 components...
Training GMM for melanoma samples with K=8 components...

Training GMM for benign samples with K=9 components...
Training GMM for melanoma samples with K=9 components...

Training GMM for benign samples with K=10 components...
Training GMM for melanoma samples with K=10 components...
Opt K for benign: 4
Opt K for Melanom: 1
training done
```

PERFORMANCE METRICS



Overall Accuracy: 86%

Threshold (0.5)

Benign Classification

Precision: 0.90

Recall: 0.95

F1-Score: 0.92

Strong performance (6231 benign vs 779 melanoma samples).

Melanoma Detection

Precision: 0.23

Recall: 0.12

F1-Score: 0.16

Only 12% of actual melanomas detected. High false negative rate!!

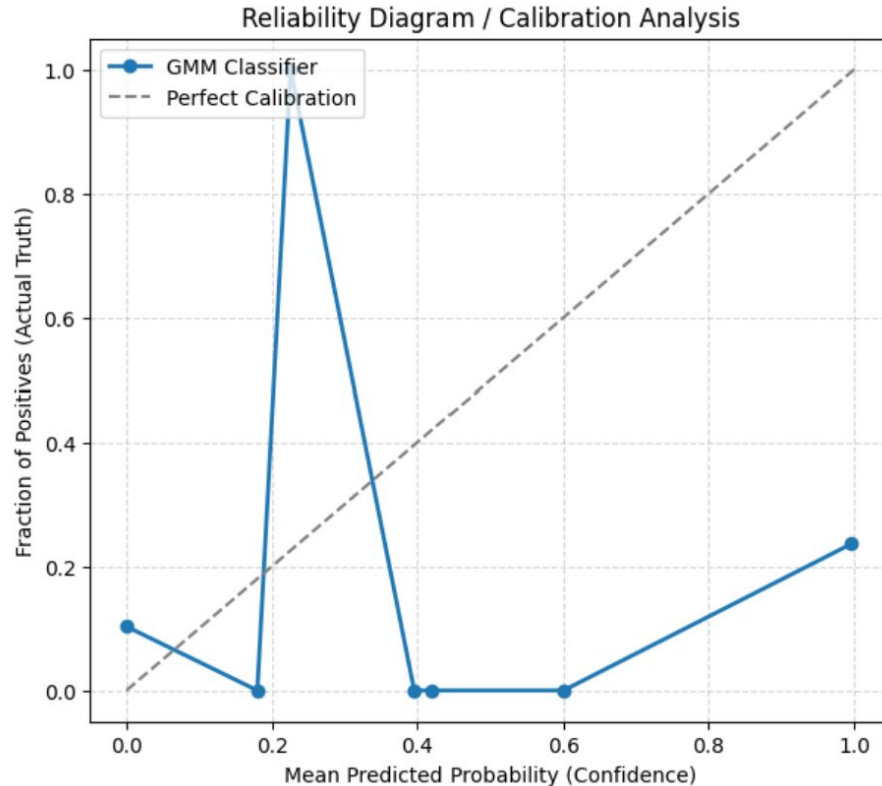
UNCERTAINTY & RELIABILITY ANALYSIS

- Severe Miscalibration

The GMM Classifier (blue line) significantly deviates from the ideal dashed line, indicating its confidence scores don't reflect true clinical probabilities.

- The Danger Zone (Overconfidence)

With confidence scores between 0.4 and 0.6, the actual rate of positive melanomas is 0.0, suggesting blind guesses without alignment.





THANK YOU

Topic B: Bayesian Decision and Uncertainty: Skin Lesion Triage

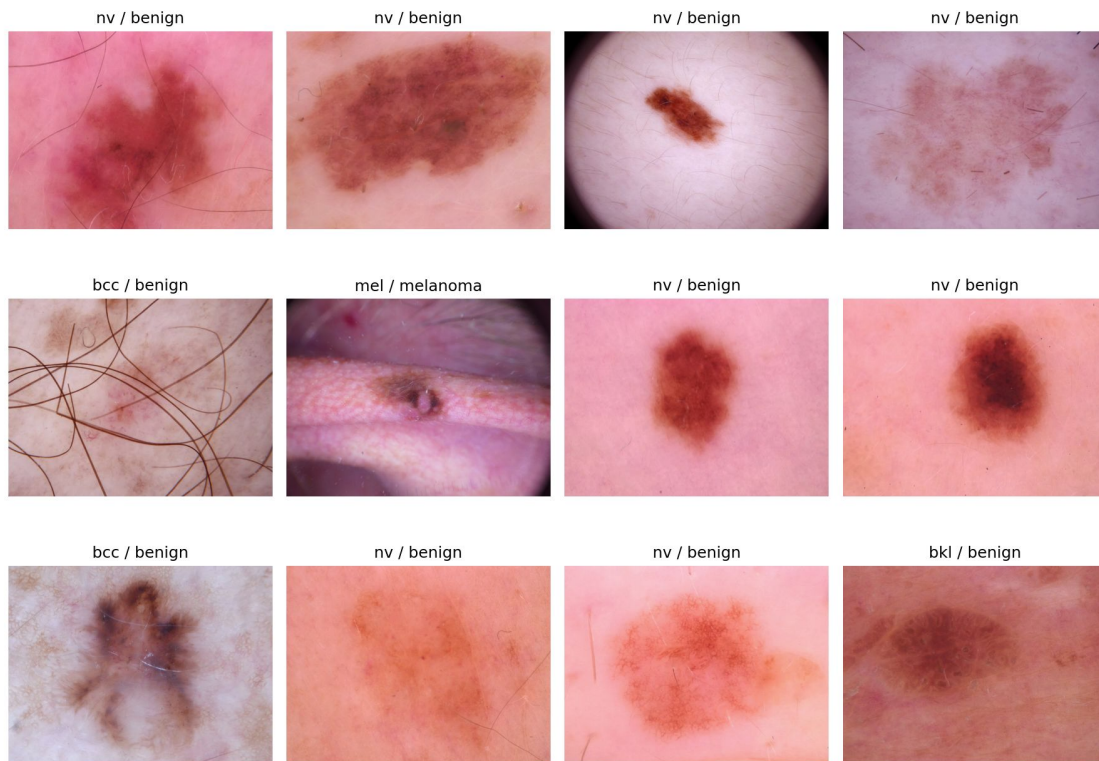
Task:

- HAM10000 dermoscopy images
- Binary triage: melanoma vs benign
- Class imbalance: melanoma is the minority class

First presentation: classical implementation
plan + initial baselines

Group: 17

Natalie Seiffert
Finn Frederick Hans Watermann
Maximilian Johannes Winkler
Ekaterina Zhittsova



Example HAM10000 images with original and binary labels

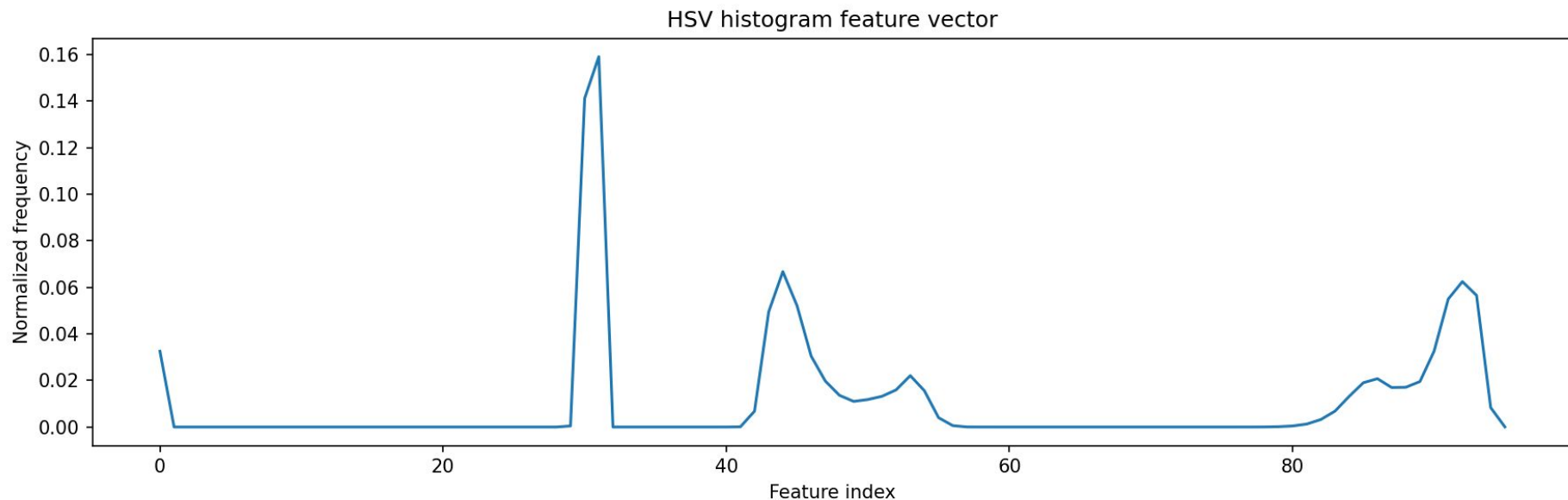
Classical Pipeline: HSV Histogram Features

Dataset/ label:

- HAM10000 dermoscopy images with diagnostic labels.
- Binary target: mel = melanoma; all other diagnoses = benign/non-melanoma.
- Strong class imbalance: melanoma is the minority class.

HSV histogram feature

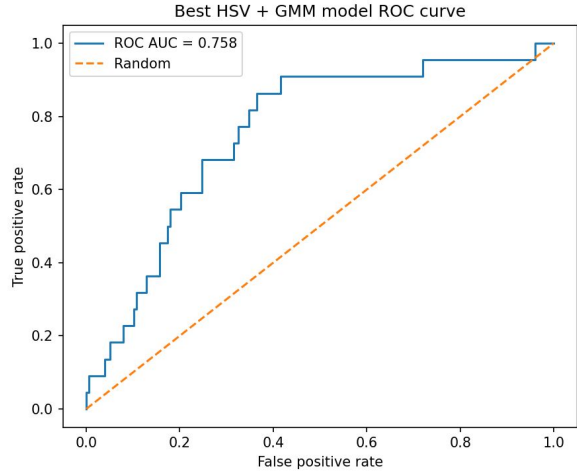
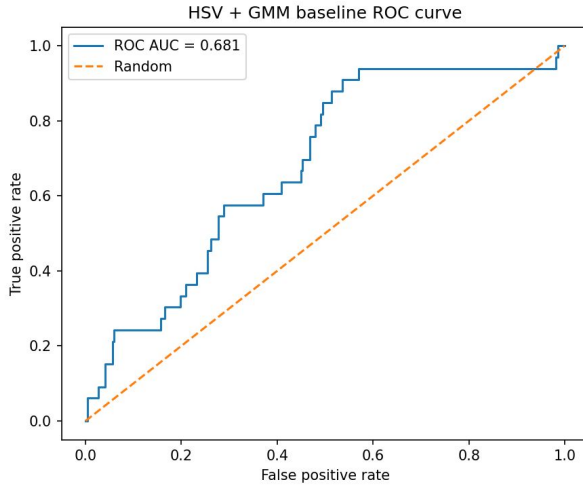
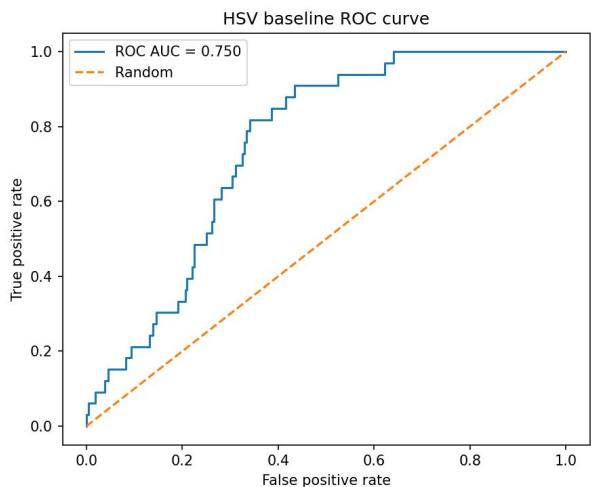
- Convert image from RGB to HSV colour space.
- Compute 32-bin histograms for H, S and V channels.
- Concatenate into a 96-dimensional feature vector.



Baseline Models: HSV Logistic Regression and HSV + GMM

Current baseline results

Model	Feature	ROC - AUC	Comment
HSV + Logistic Regression	HSV histogram, 96 features, balanced class weights	0.750	Simple discriminative baseline
HSV + GMM	One GMM for benign, one for melanoma	0.681	Generative probabilistic baseline
Best HSV + GMM variant	BIC/model search + PCA; evaluated on test set	0.758	Best current classical result

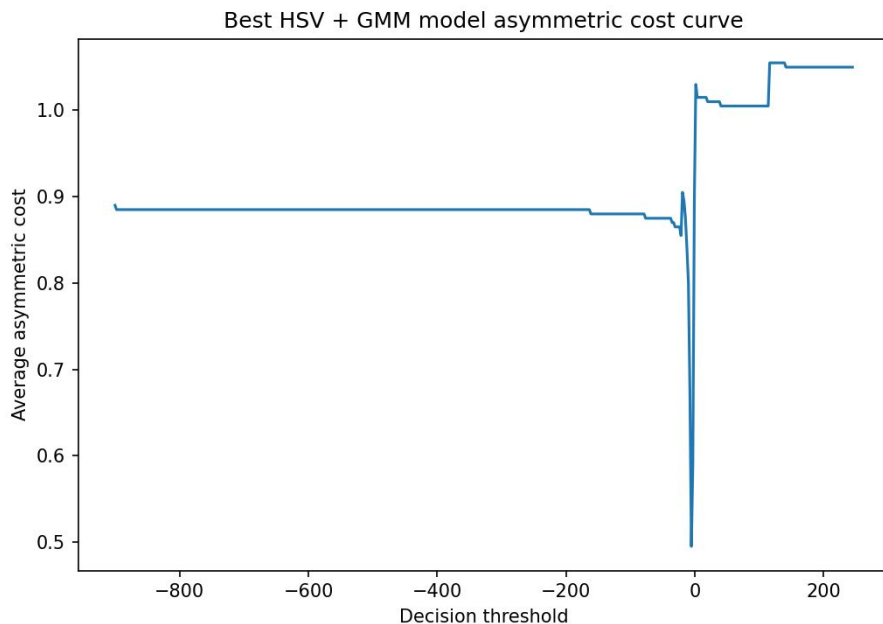


ROC curves summarize ranking performance.

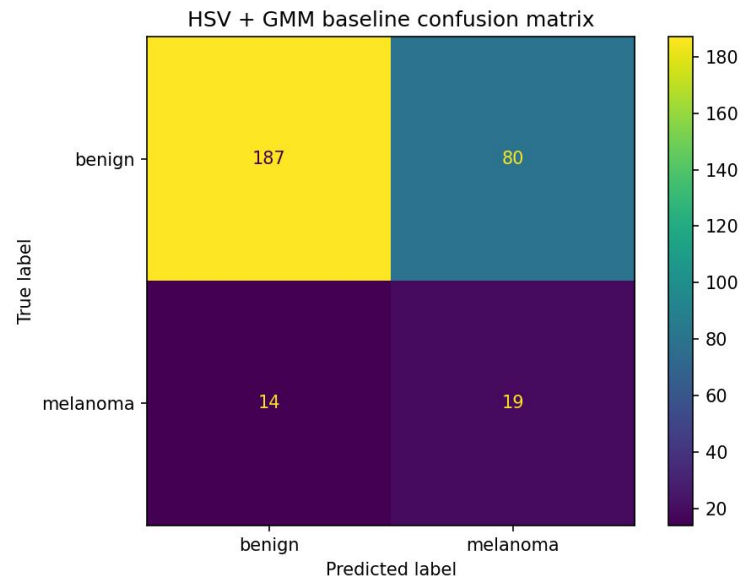
Bayesian Decision Rule, Cost Matrix

Asymmetric-cost decision idea

- False negatives are more costly: missing melanoma is worse than unnecessary referral
- Standard MAP decision uses threshold 0.5
- Cost-sensitive threshold should shift toward higher melanoma sensitivity



Bayesian threshold selection minimizes the expected decision cost.



Confusion matrix illustrating the trade-off between melanoma sensitivity and false-positive referrals.

Cost matrix

True benign: predict benign = 0, predict melanoma = C_{FP}

True melanoma: predict benign = C_{FN} , predict melanoma = 0

Decision threshold: $\tau = C_{FP} / (C_{FP} + C_{FN})$

(FP = false positive/ FN= false negative)

Skin Lesion Triage

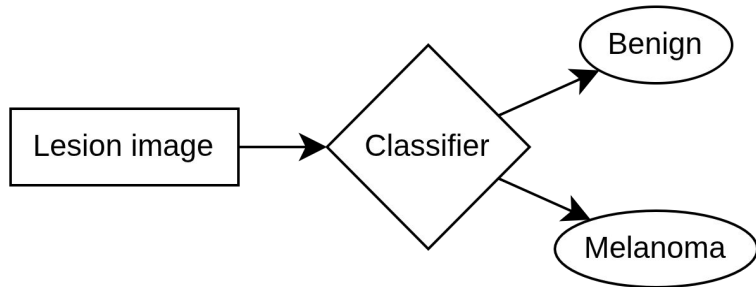
Group 42

Projects - Presentation 1

Soliman Andrawes
Anja Capo
David Martin Obina Gottschalk
Tomoki Kai Mengel

Topic

- Binary melanoma-vs-benign triage with calibrated uncertainty
 - Choosing classifiers in an environment in which false negatives are much more costly than false positives
- classical pipeline: bayesian decision theory
- modern pipeline: frozen EfficientNet-B0



Dataset and subset

- Split of underlying dataset into 70% (train), 15% (validation) and 15% (test)
 - lesion_id and image_id → put same lesion in the same subset (data leakage¹)
- validation/testing of truly new lesions rather than recognizing the same lesion

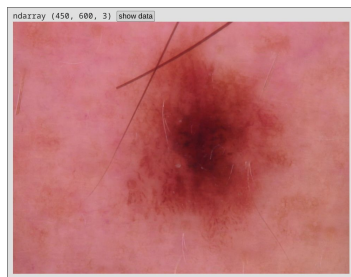


Fig. 1.: A sample lesion image

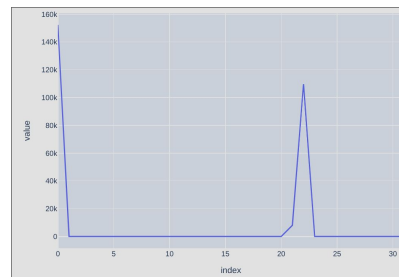
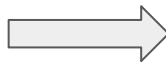


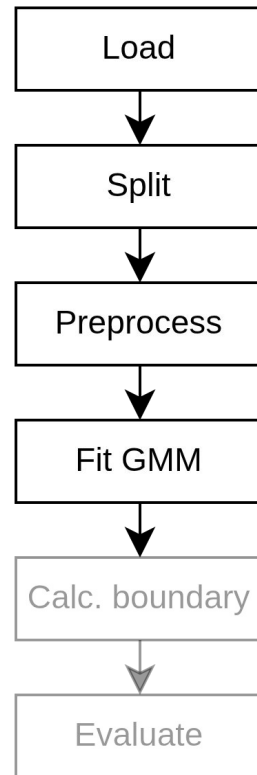
Fig. 2.: The histogram hue channel of the image in Fig. 1

Classical Pipeline

- Shrink the hsv histogram down to e.g. 32 buckets to reduce the ratio between data and dimension

→ high dimensionality leads to unstable covariance estimation

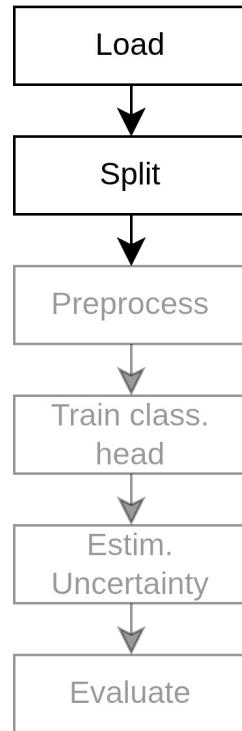
```
def process(img, histSize=32):  
    hsv_img = cv2.cvtColor(img, cv2.COLOR_BGR2HSV)  
    hist = []  
    for ch in range(3):  
        hist.append(cv2.calcHist([hsv_img], [ch], None, [histSize], [0, 255+1]))  
    return np.concatenate(hist, axis=None)
```



Modern Pipeline

- Preprocessing: Pytorch requires normalized pixels
- Transfer learning
 - EfficientNet-B0 as a feature extractor
 - Classification head
 - Standard architecture:
 - Pooling
 - Dropout
 - Dense
 - Possibly compare to other architectures
- For now we focus on MC Dropout
 - Possibly compare to other uncertainty quantifiers/estimates

Preliminary



Metrics

- confusion matrix
 - Accuracy
 - (Precision)
 - Recall
 - F1-Score
 - AUC (ROC)
 - Specificity is of high importance
- Log loss

		Predicted condition	
		Positive (PP)	Negative (PN)
Actual condition	Total population = P + N		
	Positive (P)	True positive (TP)	False negative (FN)
	Negative (N)	False positive (FP)	True negative (TN)

[2] https://en.wikipedia.org/wiki/Confusion_matrix

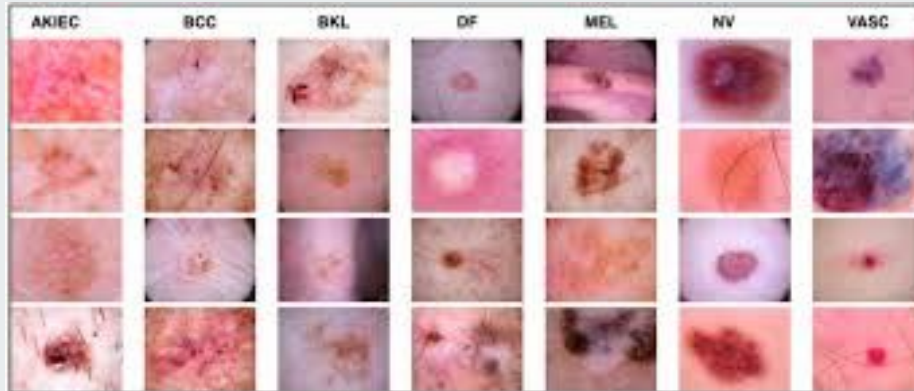
$$\text{Specificity} = \frac{\text{True Negatives}}{\text{True Negatives} + \text{False Positives}}$$

[3] <https://www.geeksforgeeks.org/machine-learning/difference-between-sensitivity-and-specificity/>

Automatic Image Analysis

Topic B: Bayesian Decision and Uncertainty: Skin Lesion Triage

Presentation 1: Classical pipeline



Group 20:

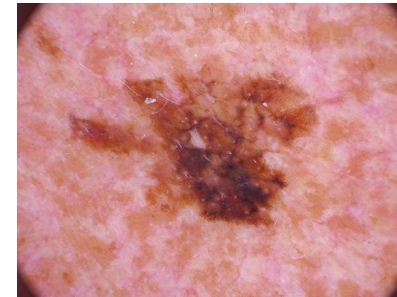
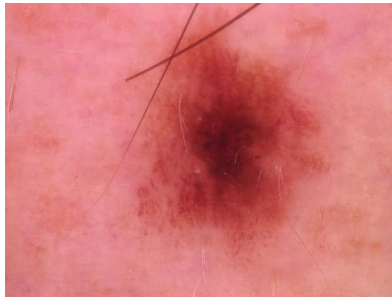
Malte Jonas Kriese
Yannik Queisler
Fabian Onur Willner
Furkan Akgun



Problem Statement & Project Overview

- **Task:** Binary skin lesion triage (Melanoma vs. Benign), treating all non-melanoma lesions as benign.
- **Motivation:** In clinical screening, **false negatives** are much more **costly** than **false positives**
 - **Key Question:** How should a classifier make decisions when missing a melanoma has a much higher cost?
- **Dataset Context:** HAM10000 dataset, related to the **ISIC 2018** challenge
- **Classical Approach:** Bayesian classification using HSV color histograms and class-specific Gaussian Mixture Models
- **Modern Approach (future stage):** Comparison with an EfficientNet-B0-based classifier with uncertainty estimation
- **Evaluation:** Analyze both predictive performance and decision quality using ROC analysis, calibration measures, and cost-sensitive decision boundaries

Data Set

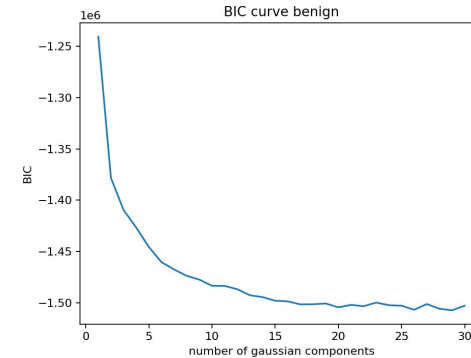
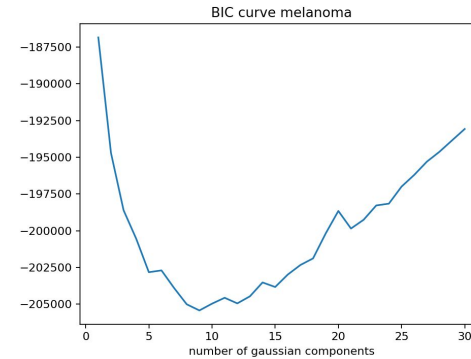


Examples of a melanocytic nevi (nv), benign keratosis-like lesions (bkl) and melanoma (mel) sample

Diagnosis	Binary role	Count	Percentage
mel	melanoma positive	1,113	12.1%
nv	benign negative	6,705	73.1%
bkl	benign negative	1,099	12.0%
df	benign negative	115	1.3%
vasc	benign negative	142	1.5%
Total	analyzed images	9,174	100.0%

Pipeline

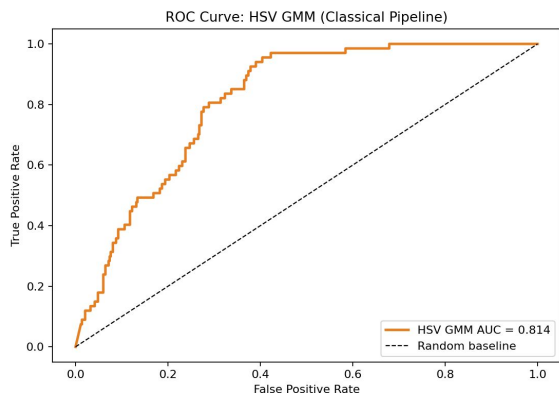
- Input gets converted to HSV
- One GMM per class gets trained
- Number of gaussian components gets selected via BIC values
- Further testing on the data to find best value



Classical Pipeline Results (HSV + GMM)

Results:

- ROC-AUC (HSV + GMM) = 0.814
- BIC-selected components: benign = 21, melanoma = 7



Discussion:

- The ROC curve is above the diagonal random baseline (AUC = 0.5), so the model learns meaningful class separation
- AUC is threshold-independent, so deployment still requires a threshold chosen for the false-negative risk.



Limitations and next steps of the Classical Approach

Limitations:

- Single split and single seed can over/underestimate true performance
- ROC-AUC = 0.814 shows good ranking, but does not guarantee low false negatives at chosen threshold

Next Steps:

- Compare against EfficientNet-B0 + MC Dropout on the same split protocol.
- Multiple seeds for inter-run variance
- Focus on reducing false negatives